Pranshul Gupta

LEAD RESEARCH DATA SCIENTIST · SAMSUNG R&D INSTITUTE INDIA - BANGALORE

(+91) 9958966084 | pranshulgupta777@gmail.com | pranshulgupta7 | pranshulgupta7

Personal Profile

AI Research and Machine Learning Leader with 7+ years of experience developing and deploying state-of-the-art NLP, Computer Vision, Recommendation Systems, and Generative AI solutions. Currently leading AI research initiatives at Samsung Research, with prior experience driving high-impact machine learning products at Paytm, Airtel, and Nagarro. Deep expertise in Large Language Models (LLMs), Transformers, Deep Learning, Representation Learning, Personalization, and Applied AI Research. Proven ability to lead multidisciplinary teams, publish research, optimize large-scale ML systems, and translate novel research ideas into production-ready solutions serving millions of users. Strong programming foundation in Python and C++ with extensive experience across PyTorch, TensorFlow, distributed computing, and modern MLOps ecosystems.

Education

Year	Degree	Institute	CPI/%
2015-2019	B.Tech Computer Science and Engineering	G.G.S.I.P.U. (MAIT)	7.85/10
2014	All India Senior School Certificate Exam(CBSE)	St. Joseph's Academy	89.6%
2012	All India Secondary School Exam(CBSE)	St. Joseph's Academy	9.6/10

Work Experience

Samsung R&D Institute India - Bangalore *Bangalore, India*

Lead Research Data Scientist, Machine Learning *July 2024 - Present*

- Led development of a **multilingual on-device notification understanding feature** that transforms mobile notifications into actionable user-intent signals to enhance **personalization** and power **Now Brief** experiences on Samsung Galaxy devices. Trained **Samsung SSCO (XLM-RoBERTa based) LLM model with LoRA adapters** on **2M+ LLM-generated and annotated samples spanning 16 languages**, achieving **94%+ F1-score** through a **hierarchical multi-level classification architecture** deployed on **Galaxy S26**.
- Architected a **hierarchical multi-level classification** framework that leveraged parent-category predictions to improve downstream intent understanding, improving fine-grained notification understanding, enabling more accurate personalization and user intent prediction across multiple taxonomy levels and powering next-generation Now Brief experiences.
- Built large-scale synthetic data generation and auto-labeling pipelines using GPT and Gemini, creating multilingual training datasets covering user activities, reservations, travel, shopping, appointments, and lifestyle events at production scale.
- Developed a **multilingual YouTube personalization feature** across **13 languages** that classifies video titles into predefined interest categories to better understand user preferences and improve personalized experiences on Samsung Galaxy devices. Utilized **Knowledge Distillation, LoRA/PEFT adaptation, and TensorFlow Lite deployment** to enable efficient on-device inference, achieving **97%+ F1-score**.
- Developed an **on-device interest classification feature** to infer user preferences from notification and engagement signals, enabling personalized experiences on Samsung Galaxy devices. Leveraged **LLM-generated training data**, **Knowledge Distillation**, and **distilled Transformer models (TinyBERT/BERT-Tiny)** to achieve **97%+ accuracy** and support next-generation personalization features on **Galaxy S26**.
- Led on-device AI optimization initiatives through Post-Training Quantization (PTQ), Quantization-Aware Training (QAT), ONNX conversion, TensorFlow Lite deployment, and Qualcomm AIMET-based Quantization-Aware Training (QAT) workflows, significantly reducing model/inference footprint while preserving production accuracy.
- Developed **Dynamic LoRA rank selection and adapter compression techniques** through custom modifications to the PEFT framework and layer-wise regularization strategies, reducing adapter size and improving deployment efficiency for resource-constrained mobile devices.
- Built an **on-device 4W (What, Who, When, Where) event extraction feature** using **Samsung Gauss LLM** to transform unstructured mobile notifications into structured event information, powering personalized **Now Brief** experiences and proactive user assistance on **Galaxy S25** devices.
- Developed a real-time **Event Classification Feature** for Samsung mobile notifications using LightGBM, achieving 98%+ accuracy and enabling accurate identification of user activities and intents to drive personalized mobile experiences.
- Designed and implemented knowledge distillation pipelines from Large Transformer-based teacher models to lightweight student architectures (TinyBERT/BERT-Tiny with custom classifier heads), achieving 92%+ accuracy while reducing latency and memory consumption for efficient on-device inference.

Paytm *Noida, India*
Senior Data Scientist *Nov. 2022 - Apr. 2024*

- Architected a **machine learning-driven advertisement ranking and campaign selection engine** that transformed billions of user interaction, transaction, and advertisement engagement signals into real-time personalization decisions across Paytm properties. Developed large-scale feature pipelines and ensemble models (**LightGBM/Random Forest, Logistic Regression**) to estimate user engagement and conversion propensity, enabling optimal ad ranking and delivering an **11% uplift in advertising revenue**.
- Developed a **brand affinity and user propensity modeling framework** to identify the top 10–20% users most likely to engage with specific brands, campaigns, and regional advertisements. Engineered large-scale behavioral features from Paytm user activity and trained **Random Forest** models to generate high-conversion audience segments, significantly improving campaign targeting efficiency and marketing ROI.
- Developed a **revenue forecasting model** for Paytm's digital advertising business using historical campaign, advertiser, and platform engagement data. Engineered time-series features and applied **ARIMA/SARIMA/SARIMAX** models to predict monthly advertising revenue, enabling data-driven business planning and revenue projections.

Bharti Airtel *Gurugram, India*
Senior Data Scientist, Machine Learning *Oct. 2021 - Nov. 2022*

- Developed a **customer acquisition and retention model** using large-scale behavioral data from Airtel's mobile application and web platforms. Leveraged **Random Forest** based propensity modeling to identify churned customers, existing subscribers, and high-value customer cohorts most likely to adopt additional Airtel products, enabling data-driven targeting and personalized outreach campaigns.

Nagarro *Gurugram, India*
Senior Data Scientist *Oct. 2020 - Oct. 2021*

- Developed an end-to-end **Question Answering pipeline** using **BERT and Transformer-based architectures**, applying transfer learning and fine-tuning techniques to support **multiple-choice, fill-in-the-blank, single-answer, and multi-answer** question formats.
- Developed an end-to-end **semantic segmentation solution** for **Maruti Suzuki** using **U-Net**, enabling automated segmentation of vehicle **doors and windows** from car images for downstream automotive applications.
- Developed an end-to-end **PCB defect detection and classification solution** using **EfficientDet** to localize and classify predefined manufacturing defects from Printed Circuit Board images, achieving **92%+ accuracy**.
- Developed lightweight **defect and object localization pipelines** using classical computer vision techniques including **SURF, FLANN, DELF, and OpenCV**, enabling model-free localization of **PCB defects** and **automotive indicator lights** for resource-constrained deployment scenarios.
- Applied **K-Means clustering** and **Principal Component Analysis (PCA)** to analyze image distributions of bridge and wall crack datasets, identifying underrepresented data segments and guiding targeted data augmentation to improve object detection model performance.

Data Scientist *July 2019 - Oct. 2020*

- Developed an end-to-end **cell nuclei detection and counting pipeline** using **Mask R-CNN based instance segmentation** to accurately identify and count individual nuclei from microscopic blood cell images, achieving **92%+ accuracy**.
- Developed **GAN-based ultrasound image generation models** to synthesize realistic medical images for healthcare imaging research and data augmentation applications.
- Developed an end-to-end **pneumonia detection solution** using **MobileNet** to classify chest X-ray images, achieving **98%+ accuracy**. Optimized the model for edge deployment through **TensorFlow Lite (TFLite)** conversion, enabling efficient on-device inference.
- Developed an end-to-end **food consumption and meal preference analytics solution** for **Japan Airlines** using **YOLO-based object detection and classification**. Analyzed meal trays to estimate dish consumption patterns, identify popular and underperforming menu items across customer segments and regions, and generate insights to improve passenger experience and menu planning. Achieved **80%+ accuracy**, and the solution was adopted by the airline prior to the COVID-19 pandemic.
- Developed a **face detection and recognition application** using **OpenCV**, enabling automated identification and verification of individuals from image and video streams.

Digital Ledeger *Gurugram, India*
Undergraduate Blockchain and MEAN Stack Developer Internship *June 2018 - July 2018*

- Developed a **decentralized pet adoption tracking application (DApp)** on the **Ethereum blockchain**, enabling secure and transparent management of adoption records through **smart contracts** without reliance on centralized intermediaries.
- Implemented **token creation, Initial Coin Offering (ICO) contracts, and smart contract interactions** using **Solidity, Truffle, and the Ethereum ecosystem**, providing cryptographically verifiable transactions and a web-based interface for blockchain operations.

Publications

Journal Articles

Trading Chatbot Using Blockchain & AI
Pranshul Gupta, Rekha Singla, Shilpa Sharma
Journal of Software Engineering And Applications. 2019

Skills

Programming	Python, C++, SQL, Data Structures & Algorithms
Machine Learning & AI	Large Language Models (LLMs), Transformers, BERT, XLM-RoBERTa, LoRA/PEFT, Knowledge Distillation, Deep Learning, Natural Language Processing (NLP), Computer Vision, Recommendation Systems, Personalization, Time-Series Forecasting
Frameworks & Libraries	PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, Pandas, NumPy, OpenCV
Model Optimization & Deployment	TensorFlow Lite (TFLite), ONNX, Post-Training Quantization (PTQ), Quantization-Aware Training (QAT), AIMET, Model Compression, On-Device AI
Software & Tools	CUDA, AWS, kafka, Airflow, Linux, Shell (Bash/Zsh), Git, Github, Jira, Microsoft Office, Google Tools.
OS and Platforms	Linux/macOS, Windows.
Languages	English (Fluent), Hindi (Native Speaker).

University Projects

Trading ChatBot Using Blockchain and AI — *MAIT, Delhi, India*
Major Project: Blockchain & AI Technology and Applications, Prof. Ms Rekha Singla, Ms Shilpi Sharma — *May 2019 - June 2018*

- Developed a **blockchain-enabled cryptocurrency trading chatbot** that combined **AI-driven trade prediction** with **smart contract based transaction management** to support automated and secure cryptocurrency trading. Designed trading strategies to analyze market trends, generate buy/sell signals, and improve decision-making while minimizing risk.
- Integrated **Blockchain 3.0 technologies** to provide transparent, tamper-resistant, and decentralized transaction processing, resulting in a secure platform for automated digital asset trading.

Insurance Management System Using Blockchain — *MAIT, Delhi, India*
Minor Project: Blockchain Technology and Applications, Prof. Ms Rekha Singla — *Nov. 2018 - Dec. 2018*

- Developed a **smart contract-based insurance management application** using blockchain technology to automate policy verification, claim processing, and participant authentication through decentralized workflows.
- Leveraged **blockchain and smart contracts** to improve transparency, reduce manual processing overhead, minimize fraud risks, and enhance trust among insurers, service providers, and policyholders.

Relevant Coursework

ONLINE COURSES

Machine Learning

Sequence Models - DeepLearning.AI, Coursera
Convolutional Neural Networks - DeepLearning.AI, Coursera
Structuring Machine Learning Projects - DeepLearning.AI, Coursera
Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization - DeepLearning.AI, Coursera
Neural Networks and Deep Learning - DeepLearning.AI, Coursera

UNDERGRADUATE

Computer Science

Data Structure & Algorithm, Advanced Algorithms, Full Stack Development, Games Development, Blockchain Technology

Awards and Achievements

2010	**Distinction**, National Talent Search Exam (NTSE)	*India*
2009	**Distinction**, International Olympiad of Mathematics	*India*
2006-2007	**Distinction**, Indira Gandhi Scholarship (IGSC) Examination	*India*
2006-2007	**Certificate**, Indian Development Foundation	*India*

Positions of Responsibility

2024-Present	**Project Lead**, Leading/Mentoring multiple Interns, Freshers and Juniors at Samsung R&D Institute India - Bangalore	*Bangalore, India*

Languages

English	Professional proficiency
Hindi	Native proficiency